File-Share & On-Prem SharePoint → SharePoint Online Migration

Implementation Solution Guide

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Executive Summary — File-Share & On-Prem SharePoint → SharePoint Online Migration

Outcome. Legacy file servers and on-premises SharePoint content are moved into SharePoint Online **cleanly** — assessed and remediated, mapped to a governed target information architecture, migrated with permissions intact, and validated with evidence — so content lands **organized and governed**, not as a cloud copy of the old mess. The result is retired file servers and on-prem farms, lower infrastructure cost and risk, and content that people can actually find and that the business can actually govern.

The problem it solves. The tempting shortcut — drag everything up with a copy tool — recreates every legacy problem in a platform where they cost more and are harder to fix: broken permissions, folders nested past the platform's limits, filenames SharePoint rejects, oversized files that fail mid-copy, and no owner for anything. Migrations that skip assessment discover these failures at scale, on cutover night, and slip by weeks.

The approach. A repeatable, **pilot-first** method using only supported Microsoft tooling (**Migration Manager** and the **SharePoint Migration Tool / SPMT**) and PowerShell:

1. **Assess** the source for the blockers that fail SPO migrations (path/URL length, invalid characters, blocked types, oversized files, permission surface).
1. **Design** a governed target IA and a permission map (source principals → SPO/M365/Entra groups), resolving every principal rather than guessing.
1. **Configure** the migration tool from the map (bulk tasks, migrate-permissions settings).
2. **Pilot** one representative scope end-to-end using **pre-stage + delta** — the bulk copy runs while users keep working, and only the changes move during a short read-only cutover window.
1. **Validate** with counts, a permission spot-check, and a delta-clean check — an explicit pass/fail gate — then **Expand** wave by wave, retiring each source only after its gate passes.

Why it's low-risk. The source stays **authoritative and read-only** — never destroyed — until each wave's validation gate passes, so rollback is simply lifting the freeze and redirecting users back. Assessment, permission-mapping, and validation are **read-only**; the only data-moving step is gated behind an explicit `-Commit`. Pre-stage + delta shrinks the downtime window from days to hours. Compliance and legal determinations (retention, legal hold, records schedules) remain with the organization's compliance function — this solution provides the technical method and the evidence, not the determination.

What's included. Solution overview, requirements guide, step-by-step implementation, testing/ validation procedures, troubleshooting, best practices, and **five PowerShell tools** — source assessment, permission mapping, task builder + runner (safe-by-default), post-migration validation, and a scope/health helper — all read-only or dry-run by default, with no credentials in code.

Bottom line. Legacy content moved to SharePoint Online the right way — assessed and remediated, mapped to a governed target IA, migrated with permissions and validated — organized, not a cloud repeat of the mess.

1. Solution Overview — File-Share & On-Prem SharePoint → SharePoint Online Migration

What this solution is

Legacy file servers and on-premises SharePoint Server farms are expensive to run, hard to secure, and increasingly a liability — but the naive fix (copy everything up with a drag-and-drop tool) simply recreates the mess in the cloud: broken permissions, orphaned folders nested twenty levels deep, filenames the platform rejects, and no owner for anything. This solution is a **repeatable, pilot-first method to migrate file shares and on-prem SharePoint content into SharePoint Online (SPO) and OneDrive cleanly** — assess and remediate the source, map it to a governed target information architecture (IA), execute with Microsoft's supported migration tooling (**Migration Manager** and the **SharePoint Migration Tool / SPMT**) using a pre-stage + delta pattern, map permissions, and validate that content and access actually landed.

It uses only supported Microsoft tooling and PowerShell (SharePoint Online Management Shell, **PnP PowerShell**, the **SPMT PowerShell** module, and Microsoft Graph). The migration-tool *console internals* (the Migration Manager agent UI, the SPMT desktop wizard screens) are deferred to Microsoft's product documentation — this solution gives you the **procedure, sequencing, and automation** around them, which is where migrations actually succeed or fail.

> ***Scope boundary.*** *This is an **on-prem / file-share** → **SharePoint Online** migration. It is distinct from any **SPO** → **other-platform** migration (e.g. SPO → Google Drive), which moves in the opposite direction with different tooling. Source here = Windows/NAS file shares (SMB/UNC) and **on-premises SharePoint Server** (2013/2016/2019/Subscription Edition) document libraries. Target = SharePoint Online team/communication sites and OneDrive for Business.*

Why this matters

- **A bad migration is worse than no migration.** Lift-and-shift with no remediation carries every

legacy problem — nested-permission sprawl, dead content, unresolvable names — into a platform where they are harder to fix and now cost you per-GB.

- **The platform will silently reject content you didn't remediate.** Path/URL length over the SPO

limit, blocked file types, invalid characters, and oversized files fail *during* the migration — discovering that on cutover night, at scale, is how migrations slip weeks.

- **Permissions are the hardest part and the part tools do worst.** NTFS ACLs and on-prem SharePoint

groups do not map one-to-one to SPO/M365 groups. Getting this wrong either over-shares (a compliance incident) or locks people out (a flood of tickets) on day one.

- **Without a target IA, you pave the cow paths.** Deciding *where* content lands — which site, which

library, which hub — before you move it is what turns a migration into a cleanup.

- **Delta migration is what makes cutover safe.** Pre-staging the bulk while the source stays live,

then moving only the changes at cutover, shrinks the freeze window from days to hours.

How it works

1. **Assess the source.** Scan file shares (and inventory on-prem SharePoint libraries) for the things

that break SPO migrations: **path/URL length** approaching the limit, **invalid characters** and reserved names, **blocked file types**, **oversized files**, very large libraries/folders, and the **permission model** (NTFS ACLs, share permissions, on-prem SP groups). The output is a factual inventory and a remediation punch-list.

1. **Remediate the source (or plan around it).** Fix or flag each blocker: shorten paths, rename

invalid items, remove or archive blocked/dead content, split oversized structures. Remediation is staged and reversible; nothing is changed in production without a report and an approval.

1. **Design the target IA.** Map each source share/library to a target **site** → **library** → **folder**

and to a hub, applying a naming convention and a governance model (owners, sharing, retention intent — with the compliance *determination* deferred to the records/legal function). This is a *mapping*, not a copy of the source tree.

1. **Map permissions.** Translate each source principal (NTFS group, on-prem SP group, individual

ACE) to a target **SPO group / M365 group / Entra security group** and a permission level. Where a clean mapping doesn't exist, it is flagged for a decision rather than guessed.

1. **Configure the migration tool.** Choose **Migration Manager** (agent-based, best for many file

shares and on-prem SP at scale, tasks defined in the SharePoint admin center) or **SPMT** (desktop / PowerShell, best for smaller or scripted runs); register the target, define tasks from the mapping (bulk CSV/JSON), and set the settings (permissions, versions, filters).

1. **Pre-stage, then delta.** Run a full **pre-stage** pass while the source stays read/write, then —

at the agreed cutover window — set the source read-only and run an **incremental (delta)** pass that moves only what changed. This is the pattern both Microsoft tools support for a low-downtime cutover.

1. **Validate.** Compare source vs. target **item counts**, spot-check **permissions**, confirm

metadata/versions per the settings chosen, and confirm the delta is clean (nothing left behind). Only then is the source decommissioned per the org's retention rules.

Reference design (what "good" looks like)

• A **pilot wave** (one representative share or on-prem site collection) proves the whole

assess → remediate → map → migrate → validate chain end-to-end before any broad wave.

• Bulk migration runs from a **migration workstation/agent** with line-of-sight to the source (SMB /

on-prem SP) and to SPO, using **Migration Manager agents** (scaled out for throughput) or **SPMT**.

• Content lands in a **governed target IA** — communication/team sites under hubs, with owners,

sharing settings, and retention intent applied — not a mirror of the old drive letters.

• Every wave uses **pre-stage + delta** so the business keeps working on the source until a short,

scheduled cutover.

• A **validation gate** (counts + permissions + delta-clean) must pass before a wave is called done and the source is retired.

Naming convention (used in every doc and script)

Every artifact this solution creates is named `<Purpose> – <Scope> – <Stage>` and reused verbatim across the docs, scripts, the SharePoint admin center, and the migration tool. The standard set:

Artifact	Name
Source assessment export	`Source Assessment – FileShare+SP – Discovery`
Remediation punch-list	`Remediation Plan – Source – PreMigration`
Target IA / destination map	`Target IA Map – SPO – Design`
Permission mapping table	`Permission Map – Source→SPO – Design`
Pilot migration task	`Migration Task – Pilot – Wave0`
First broad wave task	`Migration Task – Finance – Wave1`
Cutover runbook (per wave)	`Cutover Plan – <Scope> – <WaveN>`
Post-migration validation report	`Post-Migration Validation – <Scope> – Validate`
Target site (pilot)	`SPO Site – Pilot – Prod`

Benefits

- **Content lands organized and governed**, not a cloud copy of the legacy mess.
- **No cutover-night surprises** — blockers are found and fixed in assessment, not discovered at scale.
- **Permissions arrive intact and intentional**, mapped to modern SPO/M365 groups.
- **Low-downtime cutover** via pre-stage + delta.
- **Evidence of success** — counts and permission validation prove the migration, per wave.

Limitations (be honest with buyers)

• **Migration-tool console internals are deferred to Microsoft docs.** This solution gives the method, sequencing, task definitions, and PowerShell — not a screen-by-screen tour of the Migration Manager or SPMT UI, which Microsoft documents and changes.

• **SPO service limits are Microsoft's, not ours, and change.** Path/URL length, file-size, blocked file types, and list-view limits are enforced by the service; the exact current values must be **verified against Microsoft's documentation for your tenant/cloud** (see Requirements/Testing).

• **Government and sovereign clouds differ.** GCC High / DoD (`*.sharepoint.us` / `*.office365.us`)

and other sovereign clouds can differ in tool support, endpoints, and limits — **verify in your tenant**; do not assume Commercial behavior.

- **Not every source is tool-supported.** Encrypted, in-use, or exotic sources (e.g. some NAS permission models, IRM/RMS-protected files, third-party ECM) may need vendor tooling or manual handling; those are integrated at the pattern level and deferred to the vendor's docs.
- **Compliance/legal determinations are out of scope.** What to retain, what is under legal hold, and which records schedule applies are **decided by the org's compliance/records/legal functions** — this solution provides the technical method and evidence, not the determination.

The standard deployment model

> *Assess and remediate the source → design the target IA and permission map → configure the tool → run a **pilot** wave (pre-stage + delta) and validate → expand wave by wave, retiring each source only after its validation gate passes. Pattern: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback (the source stays authoritative until validation passes).*

2. Requirements — File-Share & On-Prem SharePoint → SharePoint Online Migration

These are the **solution-specific** prerequisites for migrating file shares and on-premises SharePoint Server content into SharePoint Online. They build **on top of** the shared `global-requirements-sharepoint.md` (M365 tenant with SPO/OneDrive and licensing; least-privilege SharePoint Administrator access; SPO Management Shell / PnP / Graph toolchain; a defined information architecture and identity/group source of truth; a pilot scope and change control; and security / compliance / records stakeholder engagement). **Verify the six shared requirements first**, then the migration-specific items below. Each item: **Description / Why / Verify / Implement if missing**.

1. Source access — file shares and/or on-prem SharePoint

- **Description:** Read access to the migration sources: file shares by **UNC path** (SMB) with an account that can read the files **and their NTFS ACLs**, and/or on-premises **SharePoint Server** (2013/2016/2019/Subscription Edition) with a farm/site-collection account that can read the content databases/libraries the migration tool will pull from.
- **Why required:** Assessment, remediation, permission mapping, and the migration tool all read directly from the source; without ACL-read rights the permission map is incomplete.
- **How to verify:** From the migration workstation, `Get-ChildItem \\server\share -Recurse` and `Get-Acl` succeed; for on-prem SP, confirm the tool's supported version and that the account can enumerate the target webs/libraries.
- **How to implement if missing:** Grant a dedicated migration service account read (and ACL-read)

on the shares and the required least-privilege role on the SharePoint farm; confirm the on-prem SP **version is supported** by the chosen tool (verify in Microsoft's Migration Manager / SPMT docs).

2. A migration workstation / agent with line-of-sight to source and SPO

• **Description:** A domain-joined Windows machine (or set of machines) that can reach the source over SMB / the on-prem SP endpoint **and** reach SharePoint Online over HTTPS (443), meeting the chosen tool's requirements — for **Migration Manager**, the **migration agent** installed and registered; for **SPMT**, the SharePoint Migration Tool and/or its `Microsoft.SharePoint.MigrationTool.PowerShell` module.

• **Why required:** The tool moves bytes through this host; throughput, reliability, and the delta passes all depend on it. Multiple agents scale throughput for large estates.

• **How to verify:** The agent shows **online/healthy** in the SharePoint admin center (Migration Manager), or `Import-Module Microsoft.SharePoint.MigrationTool.PowerShell` loads and `Register-SPMTMigration` connects (SPMT).

• **How to implement if missing:** Provision the host(s), install the agent/SPMT per Microsoft's current install docs, and allow outbound 443 to the SPO/Microsoft 365 endpoints (verify the current endpoint list for your cloud).

3. Migration tooling chosen and its version confirmed

• **Description:** A decision on the migration engine — **Migration Manager** (agent-based, tasks defined in the SharePoint admin center, strong for many file shares and on-prem SP at scale), **SPMT** (desktop/PowerShell, strong for smaller or scripted runs), or an approved **third-party** tool — with its current version installed.

• **Why required:** Task definition, CSV/bulk format, supported sources, and the pre-stage/delta behavior differ per tool; the provided `SPOMIG-TaskBuilder.ps1` targets the **SPMT PowerShell** module and emits a Migration Manager–compatible bulk CSV.

• **How to verify:** The tool launches, signs in to your tenant, and lists its supported source types and limits (confirm against Microsoft's docs).

• **How to implement if missing:** Install the current tool; if using a third-party engine, its console is deferred to the vendor's docs — this solution still supplies the assessment, mapping, and validation around it.

4. Target SharePoint Online IA provisioned (or ready to provision)

• **Description:** The **target sites, libraries, and hubs** that content will land in — named per the `Target IA Map – SPO – Design` — provisioned or scripted for provisioning, with the intended **owners** and **sharing** posture set.

• **Why required:** Migration tasks need a destination URL; migrating into an ungoverned or

not-yet-created site is how content lands orphaned.

• **How to verify:** The target site URLs resolve and the migration account can write to them (`Connect-PnPOnline` + `Get-PnPWeb` succeed on the destination).

• **How to implement if missing:** Provision the target IA first (often the *SharePoint Online Architecture & Site Provisioning* solution); at minimum create the pilot target `SPO Site – Pilot – Prod` before the pilot wave.

5. Sufficient SharePoint Online storage and a checked list of service limits

• **Description:** Enough **tenant + per-site storage** for the migrated content, and a documented check of the current SPO **migration-relevant service limits**: URL/path length, file-size limit, invalid characters / reserved names, blocked file types, and per-library/list-view thresholds.

• **Why required:** Insufficient storage stalls a wave; unremediated limit violations fail items mid-migration. These limits are **Microsoft's and can change** — they must be read from current docs or measured, not assumed.

• **How to verify:** In the SharePoint admin center check tenant/site storage; confirm the current limit values in Microsoft's SharePoint limits documentation for your cloud (Commercial vs Government may differ).

• **How to implement if missing:** Increase storage / adjust quotas; feed the confirmed limit values into the assessment (`SPOMIG-Assessment.ps1` accepts them as parameters so the check tracks the real, current numbers).

6. Identity mapping — source principals resolve to Entra ID / M365 groups

• **Description:** A way to resolve source principals (NTFS local/domain groups, on-prem SP groups, individual users) to **Entra ID** users/groups, so permissions can be mapped to **SPO groups / M365 groups / Entra security groups** in the target.

• **Why required:** Permission mapping is only possible if source identities correspond to cloud identities; unmatched principals must be surfaced for a decision, not dropped.

• **How to verify:** Users/groups from the source ACLs exist in Entra ID (spot-check with Graph or `Get-PnPAzureADGroup` / `Get-MgUser`); confirm whether **Entra Connect** hybrid sync is in place.

• **How to implement if missing:** Ensure directory sync / identity provisioning is complete for the in-scope population before the permission map is finalized; `SPOMIG-PermissionMap.ps1` reports unmatched principals so gaps are explicit.

7. A cutover window and change control for each wave

• **Description:** An agreed **cutover window** per wave during which the source can be set

read-only, the delta pass runs, and users are redirected — under the change-control process from the shared requirements.

• **Why required:** The pre-stage/delta pattern needs a defined moment to freeze the source; migration cutover is **high-impact** (users switch to a new location) and needs approval and a rollback.

• **How to verify:** A named change window, an approver, a user-communication plan, and a rollback (source stays authoritative until validation passes) are documented in the `Cutover Plan` for the wave.

• **How to implement if missing:** Schedule the window, draft the comms, and document the rollback before the wave's delta/cutover step.

*All six shared SharePoint requirements plus these seven migration-specific items should be green before the pilot wave. The migration follows the house model — **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** — with the source remaining authoritative until each wave's validation gate passes. Automation is **PowerShell** with **no hardcoded credentials or secrets**.*

Global Requirements — SharePoint Online & OneDrive Solutions

These requirements apply to **every** solution in the **SharePoint** category (SharePoint Online architecture & provisioning, external sharing, OneDrive for Business, migration to SPO, permissions, search, and records/retention). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every SharePoint solution).*** *The **SharePoint admin center**, the **Microsoft 365 admin center**, the **SharePoint Online Management Shell** and **PnP PowerShell**, and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Where a capability, setting, or portal path is **plan- or version-dependent** (storage limits, sharing controls, Syntex/records features, Commercial vs Government cloud), it is **cited or flagged "verify in your tenant"** — never invented. **Third-party migration and governance tools** (e.g. ShareGate, Migration Manager, AvePoint) are integrated at the pattern level; their consoles are deferred to the vendor's docs. Automation is **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with SharePoint Online / OneDrive and licensing

• **Description:** A Microsoft 365 tenant with **SharePoint Online** and **OneDrive for Business** and the licensing the capability requires (e.g. an E-plan for advanced records/retention, Syntex where used).

• **Why required:** All site, sharing, OneDrive, and records configuration depends on the right plan.

• **How to verify:** In the M365 admin center confirm the SharePoint/OneDrive subscription and add-ons.

• **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Administrative access (least-privilege SharePoint roles)

• **Description:** Access with the least-privilege role the solution names — typically **SharePoint

Administrator** (and Global Administrator only where genuinely required, e.g. tenant-wide sharing).

- **Why required:** Tenant/site settings, sharing, provisioning, and records changes require the role.
- **How to verify:** Confirm the role in the M365 admin center / Entra roles.
- **How to implement if missing:** Have a Global Admin assign the least-privilege role under change control.

3. SharePoint Online Management Shell / PnP + Graph toolchain

- **Description:** A workstation with the **SharePoint Online Management Shell** and/or **PnP

PowerShell** (and Graph PowerShell where used), modern-auth / MFA-capable admin sign-in (PnP app registration where required).

- **Why required:** Configuration, automation, and the provided scripts run through SPO/PnP/Graph.
- **How to verify:** `Connect-SPOService` / `Connect-PnPOnline` succeeds; modules installed.
- **How to implement if missing:** Install the modules; register the PnP app where needed; confirm sign-in.

4. Information architecture & identity/group source of truth

- **Description:** A defined **information-architecture** intent (hub/site structure, taxonomy) and the

identity/group source (Entra ID groups, M365 Groups backing team sites) that drives access.

- **Why required:** Provisioning, permissions, search, and governance key off the IA and groups.
- **How to verify:** Confirm the intended hub/site structure and which groups model is authoritative.
- **How to implement if missing:** Define the IA and group model (often the Architecture solution) first.

5. A pilot scope and change control

- **Description:** A **pilot site collection / hub / user set** to validate on before tenant-wide

rollout, and a **change-control** process for production changes (sharing, permissions, retention, migration cutover are high-impact).

- **Why required:** Every solution follows an **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**

model; tenant-wide sharing/permission/retention changes are high-impact.

- **How to verify:** Confirm a named pilot scope and an approval/change window are available.
- **How to implement if missing:** Create a pilot scope and agree change control before production changes.

6. Security, compliance & records stakeholders

- **Description:** Engagement with the **security, compliance/records, and data-governance** stakeholders

whose policies the configuration must satisfy (external sharing, permissions, retention/records, sensitivity).

- **Why required:** Sharing, permission, and records settings must align to enterprise policy.
- **How to verify:** Confirm the applicable policies and named approvers.
- **How to implement if missing:** Identify the owners and capture requirements before configuring governance.

*The deployment model across every SharePoint solution is the same: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes. Automation is delivered as **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets in code.*

3. Implementation — File-Share & On-Prem SharePoint → SharePoint Online Migration

Follow these steps in order. The migration is structured **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**. Every artifact is named `<Purpose> – <Scope> – <Stage>` and reused verbatim in later steps, testing, troubleshooting, and the scripts. **The source stays authoritative until each wave's validation gate passes** — that is the rollback.

> ***Safety posture.*** *Assessment and validation are **read-only**. The remediation and the migration execution change data — those steps default to **report / dry-run** and require an explicit `-Commit` (or the equivalent approval) before anything is written or moved. Never run a broad wave before the pilot's validation gate passes.*

Phase A — Assess the source

A1. Inventory the migration sources

1. List every in-scope **file share** (by UNC path) and every in-scope **on-prem SharePoint** site collection / library. Record the owner and rough size of each.
1. Confirm source read + ACL-read access from the migration workstation (Requirements §1).

A2. Run the source assessment (read-only)

1. Run `SPOMIG-Assessment.ps1` against each source, passing the **current SPO limits** you confirmed (Requirements §5) so the check tracks Microsoft's real numbers:

```
.\SPOMIG-Assessment.ps1 -Path \\fs01\Finance -MaxUrlLength 400 -MaxFileSizeGB 250 -CsvPath .\SourceAsse
```

1. The script flags, per item: **path/URL length** near the limit, **invalid characters / reserved names, blocked file types, oversized files, deep nesting, and it summarizes the** permission surface** (distinct principals in the ACLs).
1. Save the export as `Source Assessment – FileShare+SP – Discovery`.
2. **Gate A (assessment complete):** every source has an assessment export and a blocker count. Do not

design the target until the true blocker picture is known.

Phase B — Design the target and the remediation

B1. Build the remediation punch-list

1. From `Source Assessment – FileShare+SP – Discovery`, list each blocker with its fix: shorten path,

rename invalid item, remove/archive blocked or dead content, split an oversized folder/library.

1. Record it as `Remediation Plan – Source – PreMigration`, noting which fixes happen **at source**

(rename before migrating) vs. **in-flight** (the tool can rename on ingest — confirm the tool's behavior in its docs).

1. **Compliance note:** whether legacy content is *retained, archived, or deleted* is a

compliance/records/legal determination — this plan records the technical action, not the decision to keep or destroy.

B2. Design the target IA and destination map

1. Map each source share/library to a target **site** → **library** → **folder** and a **hub**, applying the

naming convention and the intended owners and sharing posture. Do **not** mirror drive letters or deep legacy trees — flatten and organize.

1. Record it as `Target IA Map – SPO – Design` (source path → destination URL, one row per task).

This map is the input to the migration task builder.

1. Provision (or confirm) the target sites, including at minimum the pilot target **`SPO Site – Pilot – Prod`**.

B3. Build the permission map

1. Run `SPOMIG-PermissionMap.ps1` (read-only) to enumerate source principals and propose a target

mapping (NTFS/SP group → SPO group / M365 group / Entra security group + permission level):

```
.\SPOMIG-PermissionMap.ps1 -Path \\fs01\Finance -CsvPath .\PermissionMap.csv
```

1. Review every row; **resolve unmatched principals** (the script flags them) with a decision, not a

guess. Prefer mapping to **groups**, not individuals.

1. Save as `Permission Map – Source→SPO – Design`.

2. **Gate B (design complete):** remediation plan approved, target IA map complete, permission map has

zero unresolved principals.

Phase C — Configure the migration tool

C1. Verify tenant + target readiness (read-only)

1. Run `SPOMIG-ScopeHealth.ps1` to confirm the target site exists and is writable, storage is sufficient, and the migration prerequisites are green:

```
.\SPOMIG-ScopeHealth.ps1 -SiteUrl https://contoso.sharepoint.com/sites/PilotProd
```

1. Resolve anything red before defining tasks.

C2. Register the tool and define tasks from the map

1. **Migration Manager** path: in the SharePoint admin center → **Migration**, confirm the agent is **online/healthy**, then create a **bulk migration** and upload the CSV emitted from the map (see C3). *The Migration Manager console UI is deferred to Microsoft's docs.*

1. **SPMT / PowerShell** path: run `SPOMIG-TaskBuilder.ps1` to turn `Target IA Map – SPO – Design` into migration tasks. **By default it builds the bulk CSV and does a dry-run** — it only registers and starts a migration when you pass `-Commit`:

```
.\SPOMIG-TaskBuilder.ps1 -MapCsv .\TargetIAMap.csv -BulkCsvOut .\SPMT-Bulk.csv    # dry-run: build + va
```

1. Set the migration **settings** to match the design: **migrate permissions** (per the permission map), version history depth, filters (skip blocked/dead content), and whether the tool renames invalid items in-flight (confirm the exact settings names/behavior in the tool's current docs).

C3. Name the tasks

1. Name the pilot task `Migration Task – Pilot – Wave0` and each subsequent wave `Migration Task – <Scope> – Wave<N>` (e.g. `Migration Task – Finance – Wave1`). Reuse these names in the tool, the validation, and the cutover plan.

Phase D — Pilot wave (pre-stage → delta → validate)

D1. Pre-stage the pilot (source stays live)

1. With the source still **read/write**, run the pilot **pre-stage** pass — the full copy — for `Migration Task – Pilot – Wave0`:

- SPMT/PowerShell: `.\SPOMIG-TaskBuilder.ps1 -MapCsv .\TargetIAMap-Pilot.csv -BulkCsvOut .\SPMT-Pilot.csv -Commit`
- Migration Manager: start the pilot task from the admin center.

1. Let it complete; review the tool's per-item result report for failures and re-run/repair as needed.

Users keep working on the source — nothing has cut over yet.

D2. Cutover: freeze, delta, redirect

1. In the agreed window (`Cutover Plan – Pilot – Wave0`): set the pilot source **read-only** (remove

write on the share / set the on-prem library read-only), announce the freeze.

1. Run the **incremental (delta)** pass — it moves only what changed since pre-stage (both tools

support incremental re-runs of the same task).

1. Redirect users to the target `SPO Site – Pilot – Prod` (updated shortcuts / links / comms).

D3. Validate the pilot (read-only) — the gate

1. Run `SPOMIG-Validation.ps1` to compare source vs. target and spot-check permissions:

```
.\SPOMIG-Validation.ps1 -SourcePath \\fs01\PilotFinance -SiteUrl https://contoso.sharepoint.com/sites/P
```

1. Save as `Post-Migration Validation – Pilot – Validate`.

2. **Gate D (pilot pass/fail):**

• **PASS** if: target **item count** ≥ **source** (net of intentionally filtered content), **permission

spot-check matches **the permission map, versions/metadata match the chosen settings, and the delta report shows** nothing left behind**.

• **FAIL** if any of the above is off → **roll back** (source is still authoritative and read-only

can be lifted), fix the mapping/settings/remediation, and re-run. Do not expand on a failed pilot.

Phase E — Expand wave by wave

E1. Repeat the proven chain per wave

1. For each subsequent scope (e.g. `Migration Task – Finance – Wave1`), reuse the exact pattern:

assess → remediate → confirm map → pre-stage (source live) → delta cutover (source read-only) → validate.

1. Save each wave's evidence as `Post-Migration Validation – <Scope> – Wave<N>` and pass its gate

before retiring that source.

E2. Retire the source (only after the gate)

1. After a wave's validation **passes** and a defined soak period elapses, decommission the source per

the org's **retention rules** — the *decision* to delete or archive is the compliance/records function's, executed here as the technical step.

1. Update the migration tracker: waves complete, sources retired, validation evidence filed.

Rollback (every wave)

Because pre-stage runs while the source is live and the source is only ever set **read-only** at cutover (never deleted at cutover), rollback is: **lift read-only on the source, redirect users back, and re-plan.** The source remains the authoritative copy until that wave's validation gate passes and the soak period ends. No source is destroyed on the strength of an unvalidated migration.

5. Testing & Validation — File-Share & On-Prem SharePoint → SharePoint Online Migration

Each test states **how to run it** and the **expected pass/fail signal**. Assessment and validation are **read-only**. Run the pilot-scope tests before any broad wave; re-run the per-wave tests for every subsequent wave. The overall gate is: **content migrated (counts), permissions migrated (spot-check), delta clean, target governed.**

Test 1 — Assessment completeness (pre-migration)

- **Run:** `.\SPOMIG-Assessment.ps1 -Path \\fs01\PilotFinance -MaxUrlLength <current-limit> -MaxFileSizeGB <current-limit> -CsvPath .\a.csv`
- **PASS:** The export `Source Assessment – FileShare+SP – Discovery` exists and reports a blocker

count for path length, invalid characters, blocked types, and oversized files, plus a distinct- principal count for the permission surface.

- **FAIL:** The scan errors, misses subtrees (access denied), or the limit parameters don't match the

current Microsoft-documented values → fix access / confirm current limits and re-run.

Test 2 — Remediation punch-list resolves every blocker

- **Run:** Reconcile `Remediation Plan – Source – PreMigration` against the assessment export.
- **PASS:** Every blocker in the assessment has a corresponding fix (at-source rename/shorten/remove,

or an in-flight tool rename confirmed against the tool's docs). Zero blockers are unaddressed.

- **FAIL:** Any assessed blocker has no planned fix → resolve before configuring tasks (unremediated

blockers fail mid-migration).

Test 3 — Permission map has no unresolved principals

- **Run:** `.\SPOMIG-PermissionMap.ps1 -Path \\fs01\PilotFinance -CsvPath .\pm.csv` then review the

`Unmatched` column.

- **PASS:** `Permission Map – Source→SPO – Design` maps every source principal to an SPO/M365/Entra

target with a permission level; **zero rows flagged unmatched**; mappings favor groups over individuals.

• **FAIL:** Any principal is unmatched or maps to an individual where a group exists → resolve with a decision before migrating (prevents day-one over-share or lockout).

Test 4 — Target readiness (scope/health)

• **Run:** `.\SPOMIG-ScopeHealth.ps1 -SiteUrl https://contoso.sharepoint.com/sites/PilotProd`

• **PASS:** Target site resolves and is writable by the migration account, storage is sufficient for the wave, and the reported prerequisites are green.

• **FAIL:** Site missing/not writable, storage short, or a prerequisite red → resolve before defining tasks.

Test 5 — Task build validates (dry-run)

• **Run:** `.\SPOMIG-TaskBuilder.ps1 -MapCsv .\TargetIAMap-Pilot.csv -BulkCsvOut .\SPMT-Pilot.csv` (no `-Commit`).

• **PASS:** A well-formed bulk CSV is produced, every row has a valid source and a resolvable destination URL, and the dry-run reports no structural errors. **No migration is started** (safe default).

• **FAIL:** Malformed rows, unresolved destinations, or the script attempts to move data without `-Commit` → fix the map / do not proceed.

Test 6 — Pre-stage pass completes

• **Run:** Execute the pilot pre-stage (`-Commit` / start the Migration Manager task) with the source **read/write**.

• **PASS:** The tool's per-item report shows all in-scope items migrated (or only expected filtered skips); failures are re-run/repaired to zero. Users are still working on the live source.

• **FAIL:** Persistent per-item failures → map to a root cause (see Troubleshooting) and repair before cutover.

Test 7 — Delta cutover is incremental and clean

• **Run:** Set the source **read-only**, run the incremental (delta) pass for the same task.

• **PASS:** The delta pass moves **only changed items** (not a full re-copy) and completes with nothing left behind; the tool reports the incremental set.

• **FAIL:** The pass re-copies everything (task identity changed) or reports items it could not carry → confirm the same task/mapping is reused and repair.

Test 8 — Item-count reconciliation (source vs target)

- **Run:** `.\SPOMIG-Validation.ps1 -SourcePath \\fs01\PilotFinance -SiteUrl <site> -Library "Documents" -CsvPath .\v.csv`
- **PASS:** Target **item count** ≥ **source count** net of intentionally filtered content; the report `Post-Migration Validation – Pilot – Validate` shows the reconciliation.
- **FAIL:** Target short of source beyond the known filter set → investigate the missing items before passing the gate.

Test 9 — Permission spot-check matches the map

- **Run:** Use the validation report's permission section (and spot-check in the SPO UI) against `Permission Map – Source→SPO – Design` for a sample of libraries/folders.
- **PASS:** Sampled target permissions match the mapped groups and levels — no unexpected broad access,

no missing access for a mapped group.

- **FAIL:** Over-share (e.g. "Everyone"/broad group where the map said a scoped group) or a mapped group missing → remediate permissions before declaring the wave done.

Test 10 — Target governance applied

- **Run:** Confirm the target site/library has the intended **owner**, **sharing** posture, and any **retention intent** label from the design (compliance *determination* deferred to records/legal).
- **PASS:** Owner set, sharing matches design, retention intent applied where required.
- **FAIL:** Orphaned site (no owner), sharing looser than design, or missing retention intent → apply before retiring the source.

The wave gate (all must pass to expand / retire the source)

1. **Test 8** — counts reconcile.
2. **Test 9** — permissions match the map.
3. **Test 7** — delta clean, nothing left behind.
4. **Test 10** — target governed.

If any fail, **roll back** (source is still authoritative; lift read-only) and re-run. Do not expand on a failed pilot, and do not retire a source until its wave gate passes and the soak period ends.

6. Troubleshooting — File-Share & On-Prem SharePoint → SharePoint Online Migration

Common failure modes and how to resolve them. Symptoms are grouped by phase. Where a limit or behavior is Microsoft's and version/cloud-dependent, **verify the current value in Microsoft's docs for your tenant/cloud** rather than assuming.

Assessment & remediation

Items fail migration for "path too long" even though the source path looked fine

- **Cause:** The failure is the **projected SPO URL** length (target site URL + relative path, with spaces URL-encoded), not the raw source path — the destination base adds length.
- **Resolve:** Re-run `SPOMIG-Assessment.ps1` with the **real target base URL** (`-TargetBaseUrl`) and the **current** `-MaxUrlLength`; shorten deep folders or land the content under a shorter target path. Flatten legacy tree depth in the `Target IA Map`.

"Invalid character" / "reserved name" failures

- **Cause:** Names containing SPO-invalid characters, trailing spaces/dots, or reserved leaf names.
- **Resolve:** Rename at source per `Remediation Plan – Source – PreMigration`, or confirm the tool's **rename-on-ingest** behavior in its current docs and enable it. Don't assume the tool auto-fixes — verify.

Assessment misses subtrees ("access denied")

- **Cause:** The migration account lacks read/ACL-read on some folders.
- **Resolve:** Grant the account read + ACL-read across the whole in-scope tree (Requirements §1) and re-run; an incomplete assessment yields an incomplete permission map.

Permission mapping

`SPOMIG-PermissionMap.ps1` *reports unmatched principals*

- **Cause:** Source NTFS/SP principals (local groups, deleted accounts, un-synced identities) don't resolve to Entra ID objects.
- **Resolve:** Complete directory sync for the population, replace local/stale groups with Entra groups, and make a **decision** for each unmatched principal — never migrate it as a guess. The wave gate requires **zero** unmatched.

Day-one over-share or lockout after cutover

- **Cause:** A principal mapped to too broad a group (over-share) or a mapped group missing on the target (lockout).
- **Resolve:** Re-check `Permission Map – Source→SPO – Design`, prefer scoped **groups** over "Everyone"

/ broad groups, re-apply, and re-run the validation permission spot-check (Testing §9).

Migration execution (Migration Manager / SPMT)

The delta pass re-copies everything instead of only changes

- **Cause:** The incremental pass isn't the **same task/mapping** as the pre-stage, so the tool treats

it as a new full migration.

- **Resolve:** Reuse the identical task (`Migration Task – <Scope> – Wave<N>`) and destination for the

delta; incremental only works against the same task identity.

`SPOMIG-TaskBuilder.ps1` *moved nothing / built only a CSV*

- **Cause:** That is the **safe default** — it dry-runs (build + validate) and does not start a

migration without `-Commit`.

- **Resolve:** After validating the CSV, re-run with `-Commit` to `Register-SPMTMigration` /

`Start-SPMTMigration`. If it still won't start, confirm the `Microsoft.SharePoint.MigrationTool.PowerShell` module is installed and sign-in succeeded.

Migration Manager agent shows offline / tasks stuck

- **Cause:** The agent host lost line-of-sight to the source (SMB / on-prem SP) or to SPO (443), or the

agent service stopped.

- **Resolve:** Confirm the agent is **online/healthy** in the SharePoint admin center, the host can

reach both source and SPO endpoints (verify the current endpoint list for your cloud), and scale to **multiple agents** for throughput on large estates.

Throttling / slow throughput mid-wave

- **Cause:** SPO service throttling under heavy concurrent load, or a single overloaded agent.
- **Resolve:** Spread load across more agents, migrate during off-peak windows, and let the tool's

built-in retry/backoff run. Don't fight throttling with more parallelism on one host.

Some files skipped: blocked type / oversized / in-use

- **Cause:** Files exceeding the SPO file-size limit, on the blocked-file-type list, or locked/open at

scan time.

- **Resolve:** These should have surfaced in assessment — reconcile against

`Source Assessment – FileShare+SP – Discovery`, remediate (archive/split/close), and re-run the delta. IRM/RMS-protected or encrypted files may need vendor handling (deferred to vendor docs).

Validation & cutover

Target item count is lower than source

- **Cause:** Legitimate filtered content (system files, blocked types) **or** genuinely missed items.
- **Resolve:** `SPOMIG-Validation.ps1` reports both counts; subtract the **intended** filter set — if a gap remains, inspect the tool's per-item report for the missing items and re-run the delta before passing the gate (Testing §8).

Versions or metadata didn't come across as expected

- **Cause:** The migration **settings** (version-history depth, metadata options) didn't match the design.
- **Resolve:** Confirm the tool's settings against the design, re-run for the affected scope. Verify the exact setting names/behavior in the tool's current docs (they change).

Users still editing the source after cutover ("split-brain")

- **Cause:** The source wasn't actually set **read-only** at the delta/cutover window.
- **Resolve:** Enforce read-only on the source at cutover per `Cutover Plan – <Scope> – Wave<N>`, re-run the delta to capture stragglers, then redirect users. Keep the source read-only (not deleted) until the wave gate passes.

Government / sovereign cloud

A step or tool behaves differently in GCC High / DoD

- **Cause:** Sovereign clouds (`*.sharepoint.us` / `*.office365.us`) can differ in tool support, endpoints, and limits.
- **Resolve: Verify in your tenant** — confirm the tool supports your cloud, use the correct endpoints, and re-confirm limit values; do not assume Commercial behavior.

7. Best Practices — File-Share & On-Prem SharePoint → SharePoint Online Migration

Durable practices for migrating file shares and on-prem SharePoint into SharePoint Online cleanly.

Assess before you move — always

- **Never migrate un-assessed content.** Path/URL length, invalid characters, blocked file types, and oversized files fail *during* the migration; find them in assessment, not at scale on cutover night.
- **Track Microsoft's limits as live values, not memorized constants.** SPO service limits change and differ by cloud (Commercial vs Government / sovereign). Feed the *current* documented values into `SPOMIG-Assessment.ps1` as parameters and re-confirm per engagement.

• **Measure the permission surface early.** The number of distinct principals in the source ACLs predicts how hard the permission map will be — surface it in assessment so it isn't a surprise.

Design a target, don't mirror the source

• **Map to a governed IA; don't pave the cow paths.** Migration is the one chance to flatten deep trees, retire dead shares, and land content where it belongs. Mirroring drive letters wastes it.

• **Every target site has an owner and a sharing posture before content lands.** Migrating into an ungoverned site creates orphans on day one.

• **Name everything once** with `<Purpose> – <Scope> – <Stage>` and reuse the names across the assessment, map, tasks, validation, and cutover plan — traceability from source path to destination URL is what makes validation possible.

Permissions: map to groups, resolve every principal

• **Prefer SPO/M365/Entra groups over individuals.** Individual ACEs migrated verbatim recreate the unmanageable sprawl you're trying to escape.

• **Resolve unmatched principals with a decision, never a guess.** An unmapped principal either over-shares or locks people out — both are day-one incidents. `SPOMIG-PermissionMap.ps1` flags them so they get a decision.

• **Break-permission inheritance sparingly in the target.** Re-establish clean inheritance where the legacy source had ad-hoc unique permissions everywhere.

Migrate with pre-stage + delta — keep downtime tiny

• **Pre-stage while the source is live, delta at cutover.** Do the slow bulk copy with users still working; freeze the source read-only only for the short incremental pass. This turns a multi-day freeze into hours.

• **Reuse the same task for the delta.** Incremental only works when the delta pass is the *same* task/mapping as the pre-stage — a new task re-copies everything.

• **Scale throughput with multiple agents (Migration Manager)** for large estates rather than one overloaded host; measure and right-size.

• **Never set the source read-only before pre-stage completes** — the freeze belongs at the delta/cutover moment, not before.

Validate every wave — evidence, not optimism

• **Counts + permissions + delta-clean + governed** is the gate. A wave isn't done because the tool said "complete"; it's done when the validation report proves it.

- **Spot-check permissions in the real UI**, not just the tool's log — confirm no broad group crept in and every mapped group is present.
- **Keep the source authoritative until the gate passes and a soak period ends.** The source read-only copy is your rollback; don't destroy it early.

Safety and secrets

- **Read-only by default; writes gated behind `-Commit`.** Assessment, permission-mapping, scope-health, and validation change nothing. Task execution moves data only on explicit `-Commit`.
- **No hardcoded credentials.** Use modern-auth `Connect-PnPOnline` / `Connect-SPOService` / `Register-SPMTMigration` interactive or app-registration sign-in; never embed secrets in the scripts or the CSVs.
- **Least-privilege migration account.** Grant only the source read/ACL-read and the target contribute/owner the migration needs; don't run migrations as a tenant Global Admin by default.

Compliance stays with compliance

- **Technical method, not legal determination.** What to retain, what's under legal hold, and which records schedule applies are decided by the compliance/records/legal functions. This solution supplies the migration method and the evidence; the *decision* to delete or archive a retired source is theirs.

Government and sovereign clouds

- **Verify, don't assume.** GCC High / DoD (`*.sharepoint.us` / `*.office365.us`) and other sovereign clouds can differ in tool support, endpoints, and limits. Confirm each in your tenant before planning a wave there.

8. Deliverables Summary — File-Share & On-Prem SharePoint → SharePoint Online Migration

Everything included in this solution, on one page.

Documentation

- **01 · Solution Overview** — the migration model (assess/remediate, target IA mapping, tool options, pre-stage/delta, permissions, validation), the reference design, benefits, limitations, and the Assess → Design → Configure → Pilot → Validate → Expand deployment model.
- **02 · Requirements** — the seven migration-specific prerequisites (source access, migration

workstation/agent, tool choice, target IA, storage + service-limit check, identity mapping, cutover window) that build on the shared SharePoint global requirements.

- **03 · Implementation** — exact ordered steps: assess → remediate + design target IA + permission map → configure the tool → pilot pre-stage/delta → validate (gate) → expand wave by wave. Every artifact named; pass/fail gates and rollback included.
- **05 · Testing & Validation** — ten tests, each with pass/fail signals, culminating in the wave gate (counts + permissions + delta-clean + governed).
- **06 · Troubleshooting** — the common failure modes (blocked items, permission mismatches, delta re-copies, agent/throughput, throttling, identity gaps) and their resolutions.
- **07 · Best Practices** — durable practices for assessment, target design, permission mapping, pre-stage/delta cutover, validation, safety, and government-cloud verification.
- **Executive Summary** — outcome, approach, and why the migration is low-risk.

PowerShell tools (5) — read-only or dry-run by default, no credentials in code

1. `SPOMIG-Assessment.ps1` — *Source assessment (read-only).* Scans a file share (and inventories on-prem SP content passed in) for the blockers that fail SPO migrations — path/URL length near the limit, invalid characters/reserved names, blocked file types, oversized files, deep nesting — and summarizes the permission surface. Emits `Source Assessment – FileShare+SP – Discovery`. Current SPO limits are passed as parameters so the check tracks Microsoft's real numbers. Makes no changes.

1. `SPOMIG-PermissionMap.ps1` — *Permission-mapping helper (read-only).* Enumerates source principals (NTFS/SP groups, ACEs) and proposes a target mapping to SPO/M365/Entra groups with a permission level, flagging **unmatched** principals for a decision. Emits `Permission Map – Source→SPO – Design`. Plans only; changes nothing.

1. `SPOMIG-TaskBuilder.ps1` — *SPMT task/CSV builder + runner (safe-by-default).* Turns `Target IA Map – SPO – Design` into a migration bulk CSV and validates it (**dry-run by default**); only registers and starts the SPMT migration when `-Commit` is passed. Uses the `Microsoft.SharePoint.MigrationTool.PowerShell` module with modern-auth sign-in.

1. `SPOMIG-Validation.ps1` — *Post-migration validation (read-only).* Compares source vs. target **item counts** and spot-checks **permissions** against the permission map via PnP PowerShell. Emits `Post-Migration Validation – <Scope> – Validate` — the evidence for the wave gate.

1. `SPOMIG-ScopeHealth.ps1` — *Scope/config health helper (read-only).* Verifies the target site exists and is writable, storage is sufficient, and the migration prerequisites are green before tasks are defined. Reports only.

Named artifacts (created during implementation)

Source Assessment – FileShare+SP – Discovery · Remediation Plan – Source – PreMigration · Target IA Map – SPO – Design · Permission Map – Source→SPO – Design · Migration Task – Pilot – Wave0 · Migration Task – <Scope> – Wave<N> · Cutover Plan – <Scope> – Wave<N> · Post-Migration Validation – <Scope> – Validate · SPO Site – Pilot – Prod

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

SPOMIG-Assessment.ps1

SPOMIG-PermissionMap.ps1

SPOMIG-ScopeHealth.ps1

SPOMIG-TaskBuilder.ps1

SPOMIG-Validation.ps1